TE 357: Optical Communications

Lecture 1: Course Information & Introduction

Course Information

- **Instructor:** Prof. GADZE, James Dzisi
- **Phone:** 020 689 1515
- **E-mail:** jdgadze@gmail.com
- **Office:** Room 319, FECE Building

- **TA**: LAWERTEH, Isaiah
- **Phone: 0205631588**
- **E-mail:** isaiahanim20@gmail.com

- **TA**: ADU GYAMFI, Davida Oforiwaa
- **Phone: 0596757975**
- **E-mail:** davidaoforiwaa23@gmail.com

Introduction

- Why the need for Telecom Engineering?
- Why optical communication?
- What is optical communication: The basic blocks
- Course content

Introduction

- What need of people/Things does
  - Telecom Engineering addresses?

Introduction

- People/things have a principal need (connectivity)
  - to communicate (exchange of information)
    - This need created interest in developing communication systems

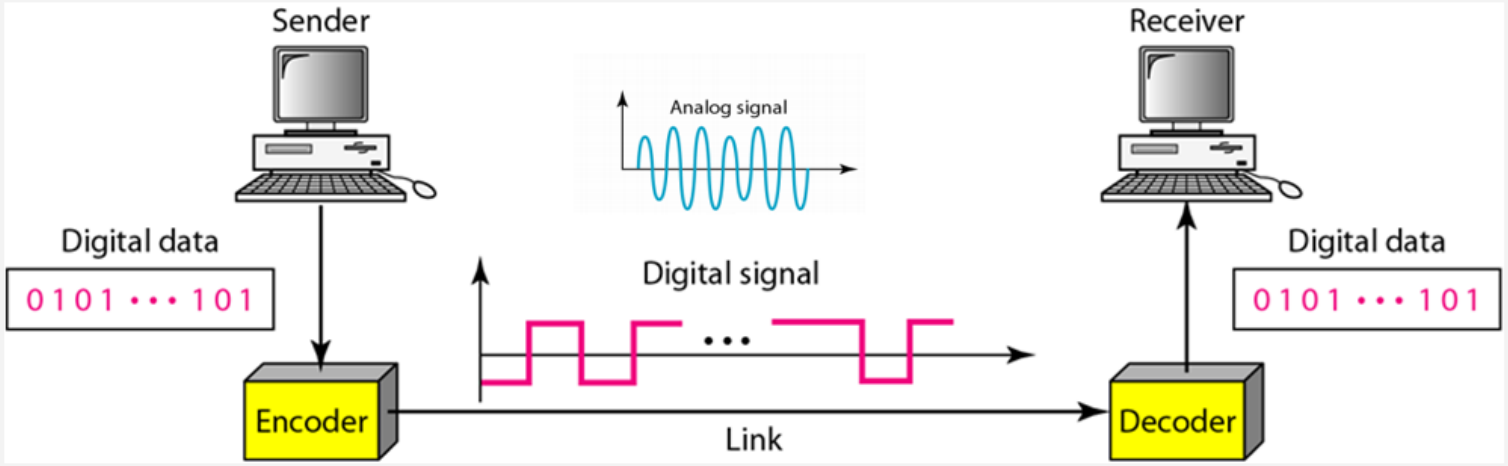
- What is a Communication system?

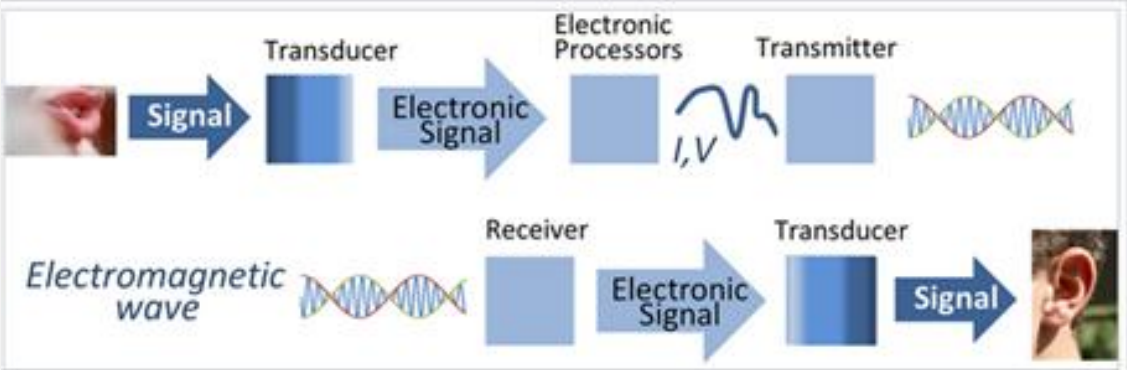
Introduction

- A Communication system is
  - a structured network of components
    - transmitter, channel, receiver that enables the transfer of information (voice, data, video) from a source to a destination

- Communication systems (Electrical/electronic or Optical)
  - Send messages from one place to another
  - Use signals to represent messages/data (TE 352)
    - Signals are in the form of electromagnetic energy

Communication systems

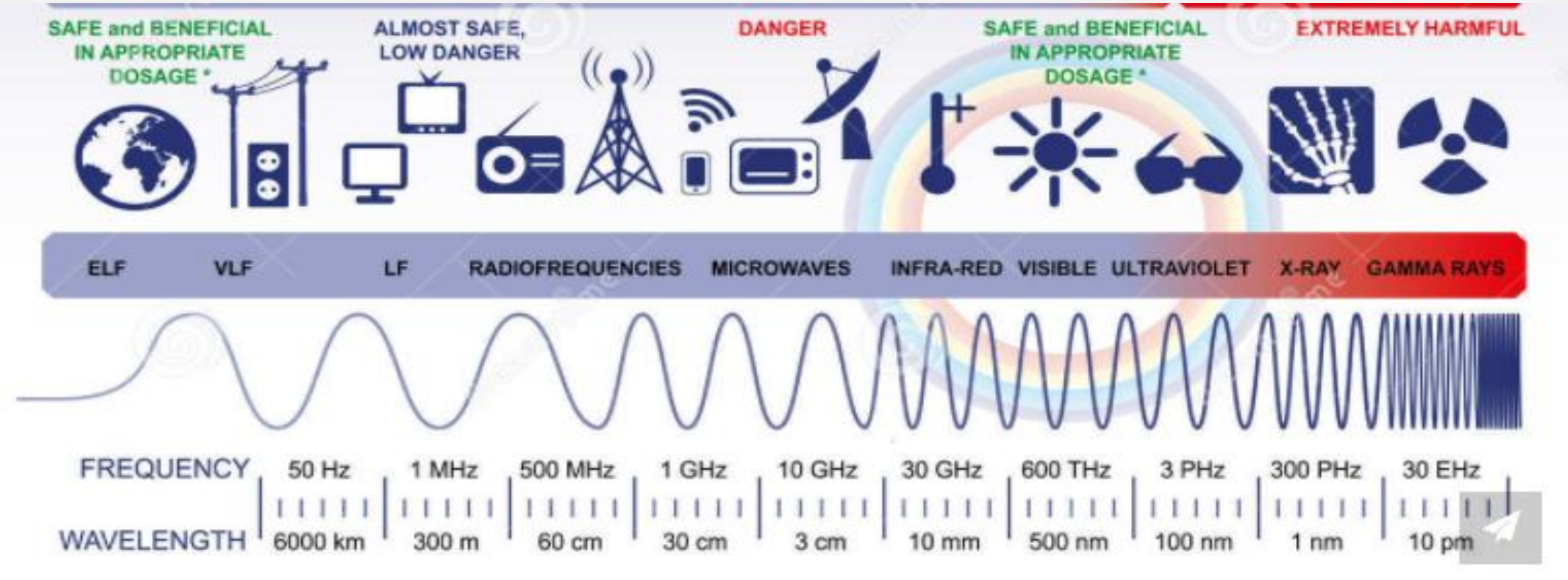




EM Spectrum

- The electromagnetic (EM) spectrum is the range of all types of EM radiation
  - Radiation is energy that travels and spreads out as it goes

EM Spectrum


SAFE and BENEFICIAL IN APPROPRIATE DOSAGE *
ALMOST SAFE, LOW DANGER
DANGER
SAFE and BENEFICIAL IN APPROPRIATE DOSAGE *
EXTREMELY HARMFUL
ELF
VLF
LF
RADIOFREQUENCIES
MICROWAVES
INFRA-RED
VISIBLE
ULTRAVIOLET
X-RAY
GAMMA RAYS
FREQUENCY
50 Hz
1 MHz
500 MHz
1 GHz
10 GHz
30 GHz
600 THz
3 PHz
300 PHz
30 EHz
WAVELENGTH
6000 km
300 m
60 cm
30 cm
3 cm
10 mm
500 nm
100 nm
1 nm
10 pm

Introduction

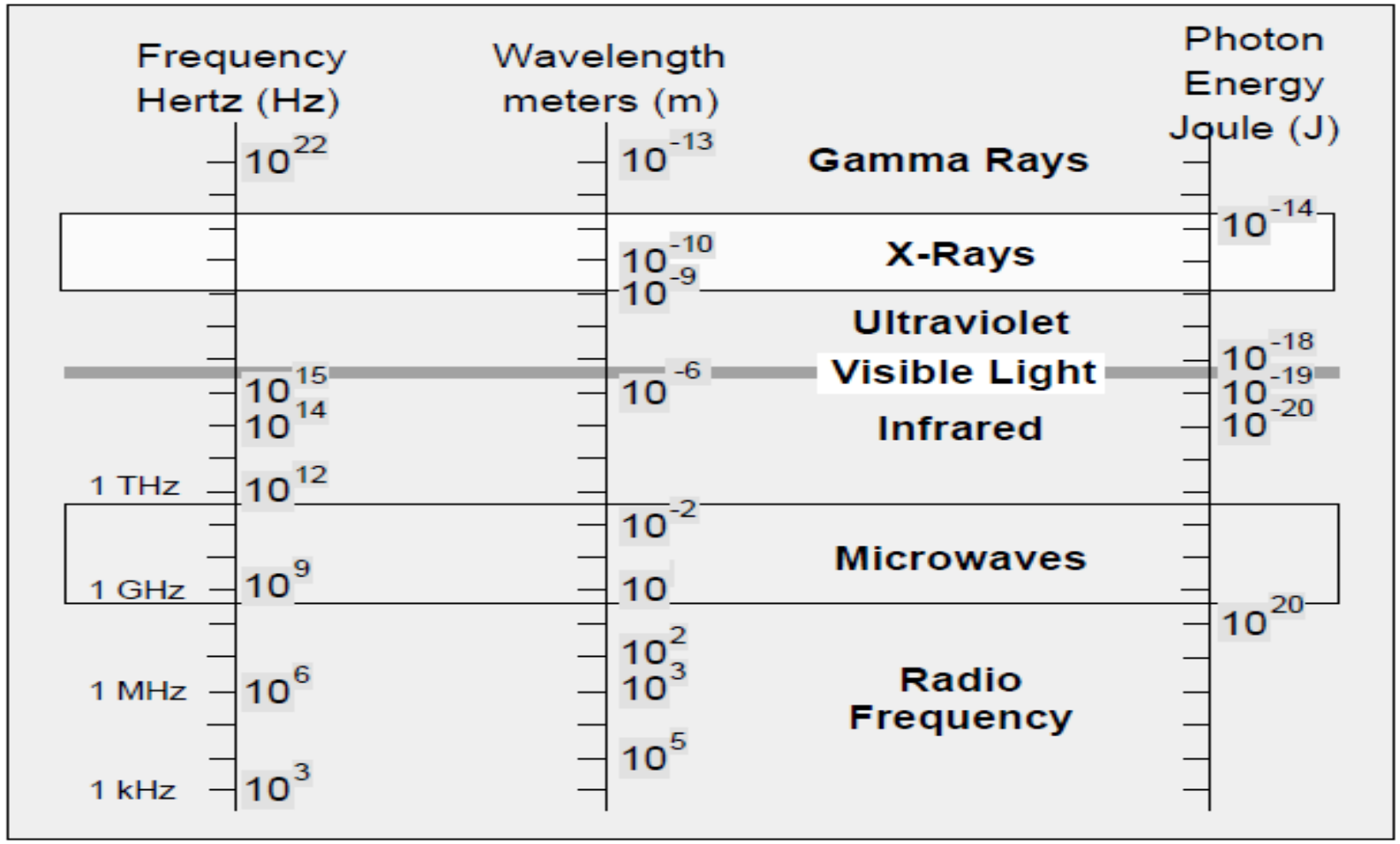
- Large proportion of
  - the electromagnetic spectrum was utilized to develop and deploy progressively more sophisticated and reliable
    - Electrical/electronic communication systems with larger capacity for conveying information
      1. Copper Cable (Wired)
      2. Coaxial (Wired)
      3. Cellular (Wireless)
      4. Microwave (Wireless)
      5. Satellite (Wireless)

Cellular

Microwave

Satellite

Electromagnetic Spectrum



Introduction

- In the past communication was confined to
  - Narrowband voice signals
    - Electronic communication
      - Voltage/current signal

Why the need for Optical Communication?

- Demand on telecommunication systems for
  - high information carrying capacity

Why the need for Optical Communication?

- Why more capacity?
  - Information age
    - Produce more information than goods
  - Use of digital transmission
    - Reliable but requires more channel capacity than analog transmission
- Though
  - Information carrying capacity is priority one for most system users
    - It cannot be increased as much as we would like

Information-Carrying Capacity

- The major limit on information-carrying capacity is
  - shown by the Shannon-Hartley theorem

$$C = B\log_2(1 + S/N)$$

- The information-carrying capacity is
  - proportional to the channel bandwidth
    - The channel bandwidth is limited by the
      - frequency of the signal carrier
        - Thus, the higher the carrier's frequency, the greater the channel bandwidth and the higher the information-carrying capacity of the system
- Rule of thumb
  - Bandwidth is approximately 10% of the carrier-signal frequency

Why Optical Communications?

- High carrier frequency: (Extremely wide bandwidth)
  - a wavelength of 1552.5 nm corresponds to a center frequency of 193.1 THz
    - Consequently, orders of magnitude increase in available transmission bandwidth & larger information capacity
- Optical Fibers
  - have small size & light weight
  - are immune to electromagnetic interference (high voltage transmission lines, radar systems, power electronic systems, airborne systems

Why Optical Communications?

- Lack of EMI cross talk between channels
- Availability of
  - very low loss Fibers (0.25 to 0.3 dB/km)
  - high performance active & passive photonic components such as tunable lasers, very sensitive photodetectors, couplers, filters
- Low cost systems for data rates in excess of Gbit/s

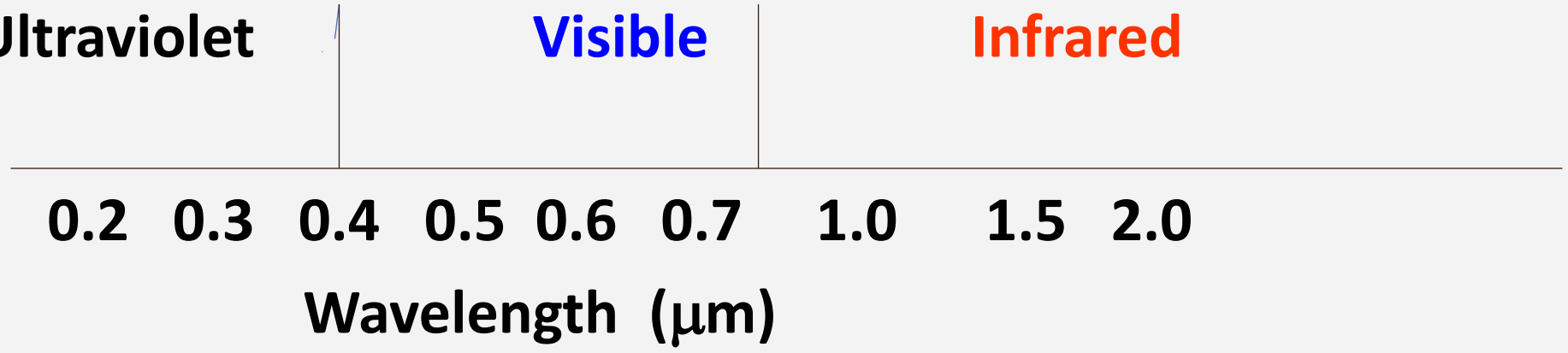
Advantages of Optical Fiber Communication

- Very high bandwidth (10 - 500 GHz, typ.)
- Very low attenuation (lowest 0.16 dB/km)
- Immune to EMI
- Data security (almost impossible to tap information)
- Lower system cost (fewer repeaters due to low attenuation of fibers)
- Small size and low weight
- Very low Bit Error Rate ($< 10^{-10}$ typically)

Optical Communication System

Most fiber optic communications systems operate in the infrared region of the spectrum

Optical Spectrum (Partial)



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